A LOCAL AI ENGINEERING PROJECT

# MeshShield

How it works, how to run it, and how to read the results

**The short version**

MeshShield generates a fictional offline-payment network, simulates delayed and repeated sync messages, builds features from past activity, and compares two small risk models with simple rules.

## Project at a glance

| Piece | What it does |
|---|---|
| Data | 1,600 reproducible, synthetic transactions (seed 2025) |
| Model | Dependency-free logistic regression with and without graph-derived features |
| Storage | Inspect-able JSON Lines datasets and a SQLite idempotency store |
| Interface | Local browser dashboard and small JSON API |
| Runtime | Python 3.11+; standard library only |

A learning and portfolio project. No payment is authorized, settled, or moved.

# 01 / THE SYSTEM

MeshShield is a research simulator that sits beside a hypothetical payment system. It receives fictional events, checks their integrity, scores risk for review, and preserves evidence about why a record was flagged.

## From offline transaction to review

| Step | What happens | Why it matters |
|---|---|---|
| 1. Create | Seeded generator creates accounts, devices, merchants, amounts, timestamps, and scenario labels. | Same seed reproduces the same demo. |
| 2. Queue | A transaction can reach the service after its event time; both event and arrival times are retained. | Delayed synchronization is visible. |
| 3. Deliver | Sync messages may be duplicated or reordered. SQLite accepts each event ID once. | Retries do not apply an event twice. |
| 4. Build features | Past account, merchant, and device activity is counted before adding the current event. | The current event cannot use its own or future history. |
| 5. Score | A logistic model scores the transaction; deterministic integrity findings are shown separately. | Risk estimates and hard checks are not conflated. |
| 6. Review | Dashboard shows holdout metrics, delivery behavior, transaction evidence, and model contributions. | A human can inspect the simulated evidence. |

**System boundary**

The local service is for a single-user demonstration. It has no authentication and binds to 127.0.0.1. It has no connection to payment rails, account providers, or real customer data.

## Project structure

src/meshshield/ separates the event contracts, generator, feature pipeline, rules, model, evaluation, simulator, storage, CLI, and web server. frontend/ contains the dashboard. docs/ explains data, features, evaluation, and safe use.

# 02 / MODEL AND EVALUATION

Two logistic-regression models use the same chronological training window. The transaction baseline uses amount, time, velocity, and sync-delay features. The graph model adds historical device/account relationship summaries. A hand-authored rule score gives a simpler third comparison.

## How to read the demo run

| Metric | Graph model | Transaction baseline | Rules only |
|---|---|---|---|
| Precision | 0.562 | 0.602 | 0.757 |
| Recall | 0.980 | 0.980 | 0.549 |
| F1 | 0.714 | 0.746 | 0.636 |
| Average precision | 0.869 | 0.888 | 0.823 |
| Brier score | 0.096 | 0.092 | 0.065 |
| Review rate @ 0.5 | 27.8% | 25.9% | 11.6% |

Measured run: 1,024 training events and 320 later holdout events, with 51 positive scenario labels in holdout. Seed: 2025.

**Interpretation**

The graph model did not improve average precision over the transaction-only baseline in this run. The rule score had higher precision at the displayed threshold but lower recall. These results show how to compare methods on this generator; they do not demonstrate real-world fraud performance.

The split is chronological: the earliest 80% trains each model and the later 20% is held out. Average precision summarizes precision across ranking thresholds. The 0.5 cut-off is illustrative, not chosen for a real analyst capacity. The Brier score is reported as probability error; calibration has not been established.

# 03 / RUN IT LOCALLY

Requires Python 3.11 or newer. The app has no external runtime dependencies. Open PowerShell in the project directory and run:

### One-time setup

```
py -3 -m venv .venv
.\.venv\Scripts\Activate.ps1
```

### Generate data, train, and evaluate

```
.\scripts\run.ps1 demo
```

### Start the dashboard

```
.\scripts\run.ps1 serve
```

Then open http://127.0.0.1:8000 in your browser.

## Useful commands

| Command | Use |
|---|---|
| .\scripts\run.ps1 generate --count 1600 --seed 2025 | Recreate the synthetic dataset and dataset card. |
| .\scripts\run.ps1 train | Fit both model variants and write holdout metrics. |
| .\scripts\run.ps1 evaluate | Print the saved evaluation results. |
| .\scripts\run.ps1 serve --port 8000 | Run the local UI and API. |

## Generated artifacts

data/generated/transactions.jsonl and sync_events.jsonl contain the synthetic records. dataset_card.json records the seed, class counts, and limitations. model.json and baseline_model.json hold model weights; metrics.json holds the evaluation. meshshield.sqlite3 persists accepted event IDs.

## Use the dashboard

- Read the comparison at the top. It is calculated from the later time holdout.
- Review the delivery summary to see duplicates ignored and invalid signatures rejected.
- Filter the table and select a transaction to inspect its scenario, integrity checks, and model contributions.
- Click Replay sync event. The API submits an event twice and reports that the retry was ignored.

Local API: GET /api/health, GET /api/metrics, GET /api/transactions?limit=100, GET /api/transactions/{id}, and POST /api/replay.

# 04 / LIMITATIONS AND NEXT STEPS

**Synthetic labels are not confirmed fraud**

The generator writes the positive label because it injected a scenario. A model can learn those artificial patterns very well and still fail on real events. All reported metrics refer only to this synthetic holdout.

## Safe interpretation

- Treat risk scores as a learning aid, never as a payment approval or decline.
- A shared device, high amount, or delayed sync can have legitimate explanations.
- Model contributions show associations in a fitted model; they do not establish cause or intent.
- The simulated signature check is a Boolean field, not cryptographic verification.
- The server has no login. Keep it bound to your own computer; do not expose it to a network.

## Possible research extensions

| Extension | Question to answer first |
|---|---|
| Scenario holdout | Does performance survive when an entire generated scenario is absent from training? |
| Review capacity | What precision and recall can be achieved for a fixed analyst review budget? |
| Calibration | Do predicted probabilities match outcome frequencies on independent labels? |
| Robustness | How does late synchronization, clock skew, or a changing device network affect results? |
| Authorized data | Can outcomes and privacy protections be independently validated before any real-data experiment? |

**More detail**

Start with README.md. The docs/ folder contains the system model, synthetic data card, feature contract, evaluation protocol, threat model, and a text walkthrough.

Built as a clean, local, reproducible portfolio project. Results are reported as measured from the generated run above.